

维修手册_TC820LTYsi_保养维护数据

维修手册知识库拆分数据 · 全字段 PDF 导出

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知识类型分布

knowledge_type	条数
maintenance_schedule	4
parameter	2
option_specification	2
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maintenance	1
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#1 lty-source-tcsi-model-specs-tc820lty			保养维护
部件	machine	章节	TCsi Specifications / Machine Specifications
起始页	24	安全等级	normal
知识类型	specification	适用机型	TC820LTYsi

摘要 | TC820LTYsi 是带 Y 轴、动力刀具和尾座销功能的型号，关键规格如下；现场维修前应核对实际配置。

A • 标识与来源

记录ID	lty-source-tcsi-model-specs-tc820lty
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	3.2.1
章节	TCsi Specifications / Machine Specifications
起始页	24
结束页	26

C • 分类与治理

记录类别	maintenance
知识类型	specification
部件	machine
安全等级	normal
可入库	是
需人工复核	是

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	TC820LTYsi 是带 Y 轴、动力刀具和尾座销功能的型号，关键规格如下；现场维修前应核对实际配置。
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关键词	· TC820LTYsi · Y轴 · 动力刀具 · 尾座销 · 规格 · 主轴
检索文本	TC820LTYsi 是带 Y 轴、动力刀具和尾座销功能的型号，关键规格如下；现场维修前应核对实际配置。；；； TC820LTYsi machine specifications: height of centers 47.3 in; maximum swing 15.75 in; swing over carriage cover 12.59 in; maximum turned length 19.29 in (varies with workholding); maximum turned diameter 8.66 in; rapid speed 787 ipm on X and Z axes and 590 ipm on Y-axis; X-axis travel 8.11 in; Y-axis travel 2.75 in; Z-axis travel 19.69 in. Headstock: A2-6 spindle nose, 7.88 in chuck diameter, M74 X 2.0 X 35 mm drawtube thread, 2.56 in bar capacity, 3.00 in spindle bore, 4.33 in spindle front bearing diameter, 4 bearings, P4 bearing class, belt drive, spindle speed 10-4000 rpm, 38.00 HP motor peak power at 480 V, 14.75 HP motor continuous power, 100 ft-lbf maximum torque at 2000 RPM, spindle motor brake, 220 ft-lbs spindle brake holding torque, 630 lbf spindle brake clamp force, 0.01 degrees spindle positioning accuracy. Tailstock: 4.72 in quill travel, 2.95 in quill diameter, MT4 taper, maximum quill thrust 1232.9 lbf at 10 kg/cm2 [142 psi]. Turret: 12 live tool stations, BMT45 type, up to 24 stations with half-index option, 0.75 in x 0.75 in turning tool shank, 1.25 in maximum boring bar diameter, live tooling maximum speed 4500 rpm, peak power 5 HP, maximum torque 38 ft-lbs. Power: 480 V, 39 A at 480 V or 90 A at 208 V with transformer option, 3 phase 60 Hz. Net size 128 in x 82 in x 79 in and 9700 lb; shipping size 138 in x 90 in x 89 in and 10700 lb. Coolant reservoir 50 gal; coolant pump delivery 5.3 GPM at 70 psi; chip wash delivery 7.9 GPM at 11.6 psi; way lube capacity 3 L; hydraulic oil capacity 11 gal; way lube oil ISO VG68 or SAE 20 W; hydraulic pump combination pump vane; chip conveyor motor 200 W; way surface hardness HRC 48-52.; TC820LTYsi; Y轴; 动力刀具; 尾座销; 规格; 主轴
原文摘录	3.0 Installation of TRAK TCsi Series Machining Centers 3.1 Installation Requirements 3.1.1 Foundation Requirements Your TCsi machine should be placed upon a solid foundation to maintain accuracy over its working life. • The TC820si machine should be placed on a solid foundation able to sustain approximately 10,000 lbs. • The TC820LTYsi machine should be placed on a solid foundation able to sustain approximately 12,000 lbs. The concrete foundation must be greater than 4” in depth and have a level and flat top surface. TRAK recommends anchoring your TCsi machine to the foundation for additional rigidity, support, and damping. If anchoring your TCsi machine to the foundation contact TRAK Customer Service at 800-367-3165 for anchoring kit and locations to add the anchors before placing the machine. Ensure the site is large enough to install and transport the whole turning center. Do not install in a location near vibration sources, such as an air compressor or punch press, etc., otherwise, poor machining accuracy may occur. 3.2 TCsi Specifications 3.2.1 Machine Specifications Capacity TC820Si TC820LTYSi Height of Centers 41.3” 47.3” Maximum Swing 15.75” 15.75” Swing Over Carriage Cover 11.81” 12.59” Maximum Turned Length (Varies With Workholding) 20.00” 19.29” Maximum Turned Diameter 8.66” 8.66” Rapid Speeds 787 ipm on Z-Axis 787 ipm on X and Z-Axis 590 ipm on X-Axis X-Axis Travel 590 ipm on Y-Axis Y-Axis Travel 6.90” 8.11” Z-Axis Travel - 2.75” 19.69” 20.8” Headstock TC820Si TC820LTYSi Spindle Nose Front A2-6 A2-6 7.88” 7.88” Chuck Diameter Drawtube Thread M74 X 2.0 X 35 mm M74 X 2.0 X 35 mm 2.56” 2.56” Bar Capacity 3.00” 3.00” Spindle Bore 4.33” 4.33” Spindle Front Bearing Diameter (ID) 4 4 Number of Bearings P4 P4 Bearing Class (Radial Runout) Drive System Belt Drive Belt Drive Spindle Speed Range 10 - 4000 rpm 10 – 4000 rpm Motor Peak Power (480V) Motor Continuous Power (480V) 38.00 HP 38.00 HP Max Torque 14.75 HP 14.75 HP Spindle Motor Brake 100 ft-lbf @ 2000 RPM 100 ft-lbf @ 2000 RPM Spindle Brake Holding Torque Spindle Brake Clamp Force - Yes Spindle Positioning Accuracy - 220 ft-lbs - - 630 lbf 0.01° Tailstock TC820Si TC820LTYSi 4.13” 4.72” Quill Travel 2.36” 2.95” MT4 MT4 Quill Diameter 690.8 lbf 1232.9 lbf Quill Taper Maximum Quill Thrust @ 10kg/cm2 [142psi] Turret TC820Si TC820LTYSi 12 Live ...

#2 lty-source-tcsi-hydraulic-pressure			保养维护
部件	hydraulic system	章节	Hydraulic Pressure
起始页	51	安全等级	high
知识类型	parameter	适用机型	TC820LTYsi

摘要 | TCsi 液压动力单元正常运行压力为 700 psi；标准三爪卡盘最大 325 psi，液压缸绝不能超过 570 psi；尾座套筒最大 140 psi。

A • 标识与来源

记录ID	lty-source-tcsi-hydraulic-pressure
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	3.12
章节	Hydraulic Pressure
起始页	51
结束页	52

C • 分类与治理

记录类别	maintenance
知识类型	parameter
部件	hydraulic system
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

检查项	· 检查液压动力单元压力表是否为 700 psi (50 kg/cm²) · 根据工件夹持装置和材料核对卡盘压力 · 根据工件夹持装置和材料核对尾座套筒压力 · 压力异常时立即停止车削中心并检查系统
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处置措施

使用机床前按夹持装置制造商规格设定压力；标准卡盘压力超过 325 psi 时用前方流量控制旋钮降低，任何情况下不得超过 570 psi；尾座套筒超过 140 psi 时降低。

适用机型

TC820LTYsi

E • 检索与原文

中文摘要	TCsi 液压动力单元正常运行压力为 700 psi；标准三爪卡盘最大 325 psi，液压缸绝不能超过 570 psi；尾座套筒最大 140 psi。
关键词	· 液压压力 · 700 psi · 325 psi · 570 psi · 140 psi · 卡盘 · 尾座套筒
检索文本	TCsi 液压动力单元正常运行压力为 700 psi；标准三爪卡盘最大 325 psi，液压缸绝不能超过 570 psi；尾座套筒最大 140 psi。；；使用机床前按夹持装置制造商规格设定压力；标准卡盘压力超过 325 psi 时用前方流量控制旋钮降低，任何情况下不得超过 570 psi；尾座套筒超过 140 psi 时降低。；The self-contained hydraulic power unit must run at 700 psi (50 kg/cm2). Stop the turning center immediately if the pressure indication is abnormal. The standard TRAK chuck maximum pressure is 325 psi (23 kg/cm2); the hydraulic cylinder is rated to 570 psi (40 kg/cm2) and chuck pressure must never exceed 570 psi. Maximum quill pressure is 140 psi (10 kg/cm2). Set chuck and quill pressure according to the workholding device and material and check the workholding manufacturer specifications.; 检查液压动力单元压力表是否为 700 psi (50 kg/cm²) ；根据工件夹持装置和材料核对卡盘压力；根据工件夹持装置和材料核对尾座套筒压力；压力异常时立即停止车削中心并检查系统；液压压力；700 psi；325 psi；570 psi；140 psi；卡盘；尾座套筒
原文摘录	3.12 Hydraulic Pressure Figure 2.12 Hydraulic Pressure Pump The self-contained hydraulic system on the TCsi machines is used to drive and control a number of functions on the machine. The hydraulic power unit located on back of the machine, needs to be running at 700psi (50kg/cm2) to provide enough pressure throughout the system, see figure 2.12 for inset of level and pressure gauges. Stop the turning center immediately if the pressure indication is abnormal. The hydraulic system is filled with ISO VG32 oil that needs replacing each year. The hydraulic system controls: • Chuck clamp and unclamp. • Turret clamp and unclamp. • Tailstock Quill forward and backward. • Tailstock body clamp and unclamp. • Tailstock Pin in and out (TC820LTY only). • Spindle Brake clamp and unclamp (TC820LTY only). • Part Catcher up and down (optional). 3.12.1 Verify Chuck Pressure Refer to figure 2.12.a the chuck pressure gauge on the front of the machine guarding. The max chuck pressure for the standard chuck installed by TRAK is 325psi (23kg/cm2). If the chuck pressure shows greater than 325psi (23kg/cm2) when the standard chuck installed by TRAK is fitted use the flow control knobs at the front of the machine to reduce the pressure as shown in figure 2.12.b. The hydraulic cylinder is rated to 570 psi (40kg/cm2), make sure the chuck pressure never goes over 570 psi (40kg/cm2). The chuck pressure should be set according to the work holding device and the material used. Please check manufacturer specs for the work holding device before use. For the standard chuck fitted by TRAK the static gripping force relationship to the chuck pressure is shown in the figures below. Please note that the gripping force shown is at 0RPM. When the spindle rotates the chuck gripping force will decrease. Figure 2.12.1a and 2.12.1b Static Gripping Force v.s. Pressure for the standard 3 jaw chuck installed on the TCsi machines. 3.12.2 Verify Quill Pressure Refer to figure 2.12.c the quill pressure gauge on the front of the machine guarding. The max quill pressure is 140psi (10kg/cm2), if the quill pressure shows greater than 140psi (10kg/cm2) use the flow control knobs at the front of the machine to reduce the pressure, see figure 2.12.c. The Quill pressure should be set according to the work holding device and the material used. Please check manufacturer specs for the work holding device before use. The tailstock quill thrust force relationship to the tailstock quill pressure is shown in the figures below. Figure 2.12.2a and 2.12.2b Tailstock Quill Thrust Force v.s. Pressure for TCsi machines. Figure 2.12.a Chuck Pressure Figure 2.12.b Chuck/Quill Figure 2.12.c Quill Pressure Decrease/Increase Pressure

#3 lty-source-tcsi-lubrication			保养维护
部件	way lubrication system	章节	Lubrication / Check and Fill Lube Tank
起始页	53	安全等级	high
知识类型	maintenance	适用机型	TC820LTYsi

摘要 | 润滑油低位报警出现后应立即补充；油箱加注清洁 ISO VG68，液位约为油箱高度的 80%，不得使用高于推荐黏度的油。

A • 标识与来源

记录ID	lty-source-tcsi-lubrication
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	3.13.2
章节	Lubrication / Check and Fill Lube Tank
起始页	53
结束页	54

C • 分类与治理

记录类别	maintenance
知识类型	maintenance
部件	way lubrication system
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

检查项	· 检查润滑油低位报警 · 检查油箱液位和润滑泵运行状态 · 检查滑轨、滚珠丝杠和运动部件是否见油 · 检查润滑管路是否断裂、堵塞或泄漏
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处置措施	取下油箱盖，加入清洁 ISO VG68 润滑油至油箱高度约 80%；报警即使断电重启仍存在，直到补油完成才会消失。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	润滑油低位报警出现后应立即补充；油箱加注清洁 ISO VG68，液位约为油箱高度的 80%，不得使用高于推荐黏度的油。
关键词	· 润滑 · 低油位 · ISO VG68 · 滚珠丝杠 · 滑轨 · 润滑泵
检索文本	润滑油低位报警出现后应立即补充；油箱加注清洁 ISO VG68，液位约为油箱高度的 80%，不得使用高于推荐黏度的油。；；；取下油箱盖，加入清洁 ISO VG68 润滑油至油箱高度约 80%；报警即使断电重启仍存在，直到补油完成才会消失。；When the low lubrication oil warning appears, check and fill the slideway lubrication system immediately. Fill the lube tank with clean ISO VG68 lubricant to approximately 80% of tank height. The warning remains after a power cycle until the tank is filled. Viscosity higher than recommended may burn out the lubrication system.；检查润滑油低位报警；检查油箱液位和润滑泵运行状态；检查滑轨、滚珠丝杠和运动部件是否见油；检查润滑管路是否断裂、堵塞或泄漏；润滑；低油位；ISO VG68；滚珠丝杠；滑轨；润滑泵

原文摘录

3.13 Lubrication The running conditions for the TCsi machines depend heavily on the lubrication management, check the lubrication system frequently to keep the machine in good condition. The lubrication pressure and frequency of distribution has been set by the manufacturer, please make sure any changes still provide adequate lubrication for the machine. It is essential that all sliding surfaces on the TCsi machines are sufficiently lubricated before starting up the machine. If the machine has just been unpacked or has not been used for a long time run the lubrication system until all the sliding parts are lubricated adequately before starting to use the machine.

3.13.1 Parts of Lubricator The following diagram explains the parts of the Ishan Precision Industries Model Yet lubrication pump. Refer to the maintenance section of this manual for frequency of service. See figure 2.13.1

1. Electrical control box cover 2. YET-C control box 3. Oil tank cap 4. Upper lid 5. Inlet filter 6. Oil tank 7. Float switch 8. Electrical control box housing 9. Pressure gauge 10. One-way elbow adapter 11. Lifting rod 12. Shaft set 13. Pressure release valve 14. Gear pump 15. Pressure adjustment knob 16. Run pump button

Figure 2.13.1 Lubricator Parts

3.13.2 Check and Fill Lube Tank A warning system is active on the control to notify users when lubrication oil is low. When the user sees this alarm message, it is essential they check and fill up the slideway lubrication system so shown in figure 2.13.2. Please fill the lube tank with oil immediately when the warning alarm message is shown. The warning alarm will remain until the tank is filled, even when the machine is power cycled. Remove the oil tank cap and fill the tank with clean ISO VG68 lubricant at the level of 80% of the tank height as shown to the right. NOTE: Viscosity higher than recommended may result in the burn out of the lubrication systems

Figure 2.13.2 Fill Tank

3.14 Installation Checklist

3.14.1 TC820si Installation Checklist Installer – Use this checklist to assure a complete setup on the TC820si.

Installation Checklist The Turning Center is packed on one wooden pallet. When receiving the machine, carefully disassemble the 1 wooden pallet and remove all the loose parts from the machine. Examine all parts of the machine to make sure that no breakage has occurred during shipping. Refer to section 2.3 of the Installation Manual.

2 During transportation, the machine body and coolant tank are packed together; first remove any brackets holding the coolant tank in place, then pull the coolant tank from the machine body and set aside. Place the forks of the fork-lift under the machine. Make sure the forks of the fork-lift protrude past the far edge of the machine base and ensure the machine is balanced before starting to lift move it. Please note that the weight of the machine is greater on the headstock side. Refer to section 2.4 of the Installation Manual.

- The machine should be lifted and moved by a fork-lift truck. Use a fork-lift truck with a minimum loading capacity of 8500 pounds to lift and move the machine.
- Pay special attention to the machine balance while lifting and moving.
- Do not stop the lifting motions suddenly during the process and prevent sudden movement as the machine could become unbalanced.
- Do not raise the machine too high as this may create a loss of stability and cause the fork-lift truck to overturn.

3 • Have another person to help guide the way while lifting and moving the machine. Ensure nobody is in the way and the path is not blocked before moving the machine.

4 Install casters on swarf conveyer while machine is lifted if the rigger has not done so already.

#4 lty-source-tcsi-diagnostic-voltage			保养维护
部件	electrical system	章节	Electrical Troubleshooting / Check Voltage
起始页	81	安全等级	high
知识类型	parameter	适用机型	TC820LTYsi

摘要 | 电气排查的关键阈值：驱动模块 480 V ±5%，两路 24 V 电源均为 24 V ±5%。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-voltage
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf

制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.8.1
章节	Electrical Troubleshooting / Check Voltage
起始页	81
结束页	81

C • 分类与治理

记录类别	maintenance
知识类型	parameter
部件	electrical system
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

故障现象	需要确认电气系统供电是否正常。
原因	输入电压或 24 V 控制电源超出允许范围。
检查项	· Combi Drive Module 输入：480 V ±5%，即 456-504 V，三相逐腿检查 · SITOP modular 10 A 输出：24 V ±5%，即 22.8-25.2 V · OMRON 200 W 电源输出：24 V ±5%，即 22.8-25.2 V
处置措施	由合格电气人员按手册和现场安全规程测量；超范围时不要继续运行。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	电气排查的关键阈值：驱动模块 480 V ±5%，两路 24 V 电源均为 24 V ±5%。
关键词	· 电压 · 480V · 24V · Combi Drive · SITOP · OMRON

检索文本	电气排查的关键阈值：驱动模块 480 V ±5%，两路 24 V 电源均为 24 V ±5%。；需要确认电气系统供电是否正常。；输入电压或 24 V 控制电源超出允许范围。；由合格电气人员按手册和现场安全规程测量；超范围时不要继续运行。；Verify Combi Drive Module input power is 480 V ±5% (456 to 504 V) across all three legs. Verify SITOP modular 10 A output is 24 V ±5% (22.8 to 25.2 V), and OMRON 200 W power supply output is 24 V ±5% (22.8 to 25.2 V).; Combi Drive Module 输入：480 V ±5%，即 456-504 V，三相逐腿检查；SITOP modular 10 A 输出：24 V ±5%，即 22.8-25.2 V；OMRON 200 W 电源输出：24 V ±5%，即 22.8-25.2 V；电压：480V；24V；Combi Drive；SITOP；OMRON
原文摘录	• Axis measurements may not repeat or be accurate. • Spindle may lose power, make a constant humming noise during operation or not turn on. Possible Cause Check This Motor drive Belt is slipping Check the alignment, condition, and tension of the Spindle speed is running at a resonance frequency Drive Belt, see Chapter 4. of the machine Speed up or slow down the spindle until vibration Defective cable connections and noise reduces Inadequate or no Lubrication Check all cable connections Make sure all the Way surfaces are getting proper Gibs too tight or loose lubrication. Check to make sure that the lube pump Calibration/backlash problem is functioning properly. Also check for any pinched or blocked oil lines. See Lubrication Chapter 2. Machine Alignment See Gib Adjustment, see Chapter 4 Reset the Backlash and check repeatability and Binding or looseness in the drive train positional accuracy. See Backlash Chapter 4. Spindle bearing noise After a crash the machine may become out of Drive, motor, or encoder failure alignment. See section 4.4. Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 Call TRAK Customer Service at (800)-367-3165 4.8 Electrical Troubleshooting 4.8.1 Check Voltage First level checking is to verify the correct supply of power at specific points. Verify power at electrical components: Verify power at input of Combi Drive Module is 480V ±5% (456 to 504 volts) (check across all 3 legs). Verify power on SITOP modular 10A power supply output is 24 volts ±5% (22.8 to 25.2 volts) (V+ and V-). Verify power on OMRON 200W power supply is 24 volts ±5% (22.8 to 25.2 volts) (V+ and V-). 4.8.2 Check Fuses Check the following fuses: Verify the fuses have the correct rating and are operational. Rated value for the following Fuses FU6 – (Barfeed Ready Option) – 3X 6A (10x38mm) FU10 – Lubrication Pump Fuse – 1X 2A (5x20mm)

#5 lty-source-tcsi-diagnostic-cables			保养维护
部件	cables and connectors	章节	Cable Connections
起始页	82	安全等级	high
知识类型	maintenance_procedure	适用机型	TC820LTYsi

摘要 | 很多“组件故障”实际来自连接不良；断电后检查清洁、重新插接并紧固所有连接器。

A • 标识与来源

记录ID	lty-source-tcsi-diagnostic-cables
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
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覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	4.11
章节	Cable Connections
起始页	82
结束页	82

C • 分类与治理

记录类别	maintenance
知识类型	maintenance_procedure
部件	cables and connectors
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

故障现象	疑似电气组件故障，但也可能是连接不良。
原因	连接器有切屑、湿气、损坏、松动或带电插拔造成损伤。
检查项	· 关机并从墙上拔掉系统电源 · 目视检查切屑、湿气和明显损坏 · 清除连接器附近切屑 · 逐一拔出并重新插回连接器 · 拧紧连接器螺钉
处置措施	禁止系统带电插拔连接器；按断电步骤重新连接后再测试。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	很多“组件故障”实际来自连接不良；断电后检查清洁、重新插接并紧固所有连接器。
关键词	· 电缆 · 连接器 · 带电插拔 · 切屑 · 湿气
检索文本	很多“组件故障”实际来自连接不良；断电后检查清洁、重新插接并紧固所有连接器。；疑似电气组件故障，但也可能是连接不良。；连接器有切屑、湿气、损坏、松动或带电插拔造成损伤。；禁止系统带电插拔连接器；按断电步骤重新连接后再测试。；A failure that appears to be an electrical component failure may be a poor connection. Turn off and unplug the system before inspecting; do not plug or unplug connectors with power on. Inspect for debris, moisture or damage, clean chips, reseal connectors one by one and tighten connector screws.；关机并从墙上拔掉系统电源；目视检查切屑、湿气和明显损坏；清除连接器附近切屑；逐一拔出并重新插回连接器；拧紧连接器螺钉；电缆；连接器；带电插拔；切屑；湿气

原文摘录

4.9 E-Stop Error (Machine Is Disabled) E stop turns power off to axis and spindle motor when the e stop is pressed a message is displayed on the control as shown in figure 3.9. To clear this message, hold down the reset button for 1 second. If this fail to clear the error, first confirm that no e stops are engaged. In the cabinet check connections to estop control relay. If these are off, then call TRAK Customer Service at (800)-367-3165. Figure 3.9 Emergency stop In the electrical cabinet check connections to E-Stop control relay. If fail to clear issue call TRAK Customer Service at (800)-367-3165. 4.10 Door Guards Not Working Properly Check the switch on top of the door. Check connections to door close relay and door release relay labeled in the electrical cabinet. If the key does not line up with the interlock, the door may have moved out of alignment. Figure 3.10 shows the alarm message when door is not secured. Figure 3.10 Door Error Alarm WARNING!!! If the door does not lock properly, do not operate machine. 4.11 Cable Connections The TCsi machines use several cables to communicate between systems. It is often the case that what appears to be the failure of an electrical component is actually attributable to a poor connection. 1. Turn off and unplug the system from the wall. WARNING! Do not plug and unplug connectors with the system power on. This may cause damage to the connector board and harm to the technician. 2. Visually inspect the connections for excessive debris, moisture, or obvious damage. 3. Carefully clean any chips away from the connectors. 4. One-by-one, take out each connector and then plug them back in. Do the same at the computer/display. 5. Make sure the screws are tightened on each of the connectors. 4.12 Chip Conveyor Troubleshooting The chip conveyor is designed to evacuate chips from the machine envelope after cutting. If not used and maintained correctly issues can occur. When the conveyor is noisy check for a buildup of chips and clear any obstructions. If noise persists check the conveyor belt for damage and call TRAK Customer Service at (800)-367-3165.

#6 lty-source-tcsi-maint-daily			保养维护
部件	machine	章节	Daily Maintenance
起始页	85	安全等级	normal
知识类型	maintenance_schedule	适用机型	TC820LTYsi

摘要 | 每日维护重点是清屑、油位、压力、润滑、卡盘和冷却液。

A • 标识与来源

记录ID	lty-source-tcsi-maint-daily
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	5.2.1
章节	Daily Maintenance

起始页	85
结束页	85

C • 分类与治理

记录类别	maintenance
知识类型	maintenance_schedule
部件	machine
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	每日维护重点是清屑、油位、压力、润滑、卡盘和冷却液。
关键词	· 日维护 · 700 psi · 70 psi · 卡盘加脂 · ISO VG68 · 冷却液
检索文本	每日维护重点是清屑、油位、压力、润滑、卡盘和冷却液。 ; ; ; ; 每日清除滑轨、护罩、卡盘、刀塔、刀架及运动部件周围大部分切屑；清空冷却液箱上方筛网；检查液压油位、700 psi 主液压压力；检查 ISO VG68/SAE 20W 润滑油液位、润滑压力 70 psi、油路和滑轨供油；检查夹紧机构并每天给卡盘加脂两次；检查冷却液液位。 ; 日维护；700 psi；70 psi；卡盘加脂；ISO VG68；冷却液
原文摘录	5.1.3 How to Replace Hydraulic Pump Filter WARNING!!! Heavy Object, us proper lifting equipment. 1. Remove the six screws and two eye bolts between the top cover and the tank, see figure 4.1.3. 2. Use proper lifting equipment to separate the top cover from the tank. 3. Lift the top cover high enough to replace the filter. 4. Replace top cover and secure to tank. Figure 4.1.3 Hydraulic Pump 5.2 Maintenance Intervals 5.2.1 Daily Maintenance • Remove majority of chips and debris from around the ways, way covers, chuck, turret, tool holders and other moving parts. • Empty chips from screen sitting above the coolant tank. • Check hydraulic oil levels. If the oil level is low check for leaks in the hydraulic system. If needed fill with ISO VG32/SAE 10W. • Check the main hydraulic pressure is set to 700psi. • Check the lubrication pump oil level and make sure it is always above the minimum line. If needed fill with ISO VG68/SAE 20W. • Check to make sure there are no obvious oil leaks to the lubrication system, and it is getting to all the slideways and moving parts. • Check lubrication pressure is set to 70psi when the pump is running. • Check to make sure the clamping mechanism is clean and in proper working order. Grease the chuck twice daily. • Check coolant level and add if low.

#7 lty-source-tcsi-maint-monthly			保养维护
部件	machine	章节	Monthly Maintenance
起始页	86	安全等级	normal
知识类型	maintenance_schedule	适用机型	TC820LTYsi

摘要 | 每月清洁电柜过滤器和滑轨；每两个月清洁冷却液箱并换液。

A • 标识与来源

记录ID	lty-source-tcsi-maint-monthly
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	5.2.2
章节	Monthly Maintenance
起始页	86
结束页	86

C • 分类与治理

记录类别	maintenance
知识类型	maintenance_schedule
部件	machine
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	每月清洁电柜过滤器和滑轨；每两个月清洁冷却液箱并换液。
关键词	· 月维护 · 空气过滤器 · 热交换器 · 冷却液箱
检索文本	每月清洁电柜过滤器和滑轨；每两个月清洁冷却液箱并换液。；；； 每月清洁电柜热交换器空气过滤器，必要时更换；拆下护罩检查滑轨并清除切屑和碎屑。每两个月排空并清洁冷却液箱，重新加入清洁冷却液。； 月维护；空气过滤器；热交换器；冷却液箱

原文摘录

5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

#8 lty-source-tcsi-maint-half-yearly			保养维护
部件	machine	章节	Half Yearly Maintenance
起始页	86	安全等级	normal
知识类型	maintenance_schedule	适用机型	TC820LTYsi

摘要 | 半年维护包括安全互锁、调平、主轴皮带、反向间隙和镶条。

A • 标识与来源

记录ID	lty-source-tcsi-maint-half-yearly
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	5.2.3
章节	Half Yearly Maintenance
起始页	86

结束页	86
C • 分类与治理	
记录类别	maintenance
知识类型	maintenance_schedule
部件	machine
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	半年维护包括安全互锁、调平、主轴皮带、反向间隙和镶条。
关键词	· 半年维护 · 互锁 · 调平 · 皮带张力 · 反向间隙 · 镶条
检索文本	半年维护包括安全互锁、调平、主轴皮带、反向间隙和镶条。；；； 每半年检查所有开关和互锁；检查机床水平并按需重新调平；检查主轴电机皮带张力并调整；检查各轴反向间隙并调整；检查各轴和尾座镶条并调整。； 半年维护； 互锁； 调平； 皮带张力； 反向间隙； 镶条
原文摘录	5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

#9 lty-source-tcsi-maint-yearly			保养维护
部件	machine	章节	Yearly Maintenance
起始页	86	安全等级	normal
知识类型	maintenance_schedule	适用机型	TC820LTYsi

摘要 | 年度维护包括液压油箱清洁换油、整机清洁、磨损间隙和线路/软管检查。

A • 标识与来源

记录ID	lty-source-tcsi-maint-yearly
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	5.2.4
章节	Yearly Maintenance
起始页	86
结束页	86

C • 分类与治理

记录类别	maintenance
知识类型	maintenance_schedule
部件	machine
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	年度维护包括液压油箱清洁换油、整机清洁、磨损间隙和线路/软管检查。
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关键词	· 年度维护 · ISO VG32 · SAE 10W · 液压油箱 · 磨损 · 软管
检索文本	年度维护包括液压油箱清洁换油、整机清洁、磨损间隙和线路/软管检查。；；； 每年拆下液压油箱、放油并清洁内部，清洁过滤器后加入 ISO VG32/SAE 10W 新液；拆下所有护罩，吸除切屑和碎屑并从上到下擦拭机床；检查异常磨损和间隙；检查电缆、冷却液软管和液压管路是否磨损或割伤。； 年度维护；ISO VG32；SAE 10W；液压油箱；磨损；软管
原文摘录	5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

#10 lty-source-tcsi-maint-gib			保养维护
部件	gibs	章节	Gib Adjustments
起始页	86	安全等级	high
知识类型	adjustment_procedure	适用机型	TC820LTYsi

摘要 | 镶条调整不当会导致机床精度损失；按 1.2 Nm 扭矩执行，不要用垫片替代正确调整。

A • 标识与来源

记录ID	lty-source-tcsi-maint-gib
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	5.3.1
章节	Gib Adjustments
起始页	86
结束页	86

C • 分类与治理

记录类别	maintenance
知识类型	adjustment_procedure
部件	gibs
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	镶条调整不当会导致机床精度损失；按 1.2 Nm 扭矩执行，不要用垫片替代正确调整。
关键词	· 镶条 · gib · 1.2 Nm · 10.62 lbf.in · 精度
检索文本	镶条调整不当会导致机床精度损失；按 1.2 Nm 扭矩执行，不要用垫片替代正确调整。；；；镶条应平整、无扭曲、无毛刺且油道无堵塞；损坏或有划痕的镶条必须更换。调整螺钉扭矩设为 1.2 Nm（10.62 lbf.in, 0.9 ft-lbs）。单面镶条需松开两颗锁紧螺母，反复移动轴并复紧；双面镶条需松开电机侧设定螺钉并按步骤复紧。；镶条；gib；1.2 Nm；10.62 lbf.in；精度

原文摘录

5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

#11 lty-source-tcsi-turret-model-difference			保养维护
部件	turret	章节	12 Station Hydraulic Turret
起始页	106	安全等级	normal
知识类型	model_difference	适用机型	TC820LTYsi

摘要 | TC820si 使用 Bolt-on 刀塔面; TC820LTYsi 使用 BMT45 刀塔面并可配动力刀具，半分度选项可提供 24 个定位位置。

A • 标识与来源

记录ID	lty-source-tcsi-turret-model-difference
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	6.1
章节	12 Station Hydraulic Turret

起始页	106
结束页	106

C • 分类与治理

记录类别	maintenance
知识类型	model_difference
部件	turret
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	TC820si 使用 Bolt-on 刀塔面；TC820LTYsi 使用 BMT45 刀塔面并可配动力刀具，半分度选项可提供 24 个定位位置。
关键词	· 刀塔 · BMT45 · Bolt-on · 动力刀具 · 24工位 · 型号差异
检索文本	TC820si 使用 Bolt-on 刀塔面；TC820LTYsi 使用 BMT45 刀塔面并可配动力刀具，半分度选项可提供 24 个定位位置。；；； The TC820si uses a bolt-on turret face. The TC820LTYsi uses a BMT45 turret face with a live tooling drive; the BMT45 turret is available with a half-index option providing 24 indexing positions. The turret uses an AC servo motor, hydraulic clamping and a proximity switch.; 刀塔; BMT45; Bolt-on; 动力刀具; 24工位; 型号差异
原文摘录	6.0 Sauter 12 Station Turret 6.1 12 Station Hydraulic Turret The TCsi series is fitted with a 12-station turret from Sauter. The TC820si uses and Bolt-on Turret face and the TC820LTYsi is fitted with a BMT45 Turret face with a live tooling drive. The BMT45 turret for the TC820LTY is available in a half index option providing 24 indexing positions. The turret is driven by an AC servo motor and utilizes a hydraulic clamping mechanism and proximity switch. As the tool exchange command is issued, turret will be rotated to the selected tool position according to NC or manual commands. When the selected tool traces the target location, the AC servo motor will stop immediately. The hydraulic system will then lock the tool magazine to prevent the tool magazine from further traveling. The proximity switch is used for checking the magazine’ s clamping. The rotation direction could be either cw or ccw. The tool is selected based on the shortest path to minimize the tool selection time. 6.2 Troubleshooting the 12 Station Turret 6.2.1 12 Station Turret Out of Sync with Control Display Occasionally the turret position will become out-of-synch with the control display. Typically, this is when the green highlighted entry of the Tool List as shown in figure 5.2.1a shows a different number than the tool position number engraved into the turret body at the current cutting position. Figure 5.2.1a Tool List Showing Tool Position To re-sync the turret with the tool table, on the MCP press “JOG” and “REF PT” . On the control press 1 - “>” arrow to pull out hidden window, 2 – turret menu, then 3 – Turret Reference as shown on figure 5.2.1b.

#12 lty-source-tcsi-high-pressure-coolant			保养维护
部件	high-pressure coolant	章节	High Pressure Coolant System
起始页	134	安全等级	high
知识类型	option_specification	适用机型	TC820LTYsi

摘要 | 高压冷却系统通过刀塔和刀具直接提供 220 psi 冷却液；压力传感器用于检测压力不足，触发时会暂停当前程序并停止机床。

A • 标识与来源

记录ID	lty-source-tcsi-high-pressure-coolant
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	9.3.1
章节	High Pressure Coolant System
起始页	134
结束页	134

C • 分类与治理

记录类别	maintenance
知识类型	option_specification
部件	high-pressure coolant
安全等级	high
可入库	是
需人工复核	否

D • 业务明细

检查项	· 检查高压泵、过滤器和压力传感器 · 压力传感器触发后清洁过滤器 · 更换传感器后设为 85 psi (6 kg/cm²) · 更换泵后确认旋向正确并至少无故障运行 10 秒
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处置措施	按断电、排液、拆卸、重新接线和泄漏检查步骤更换泵/传感器；过滤器在压力传感器触发或更换冷却液时清洁。
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	高压冷却系统通过刀塔和刀具直接提供 220 psi 冷却液；压力传感器用于检测压力不足，触发时会暂停当前程序并停止机床。
关键词	· 高压冷却 · 220 psi · 压力传感器 · 85 psi · 过滤器 · 冷却泵
检索文本	高压冷却系统通过刀塔和刀具直接提供 220 psi 冷却液；压力传感器用于检测压力不足，触发时会暂停当前程序并停止机床。；；；按断电、排液、拆卸、重新接线和泄漏检查步骤更换泵/传感器；过滤器在压力传感器触发或更换冷却液时清洁。；The high-pressure coolant system delivers coolant at 220 psi through the turret and cutting tool. The pump has a filter and pressure sensor. A low-pressure warning halts the current program and machine operation. Set a replacement pressure sensor to 85 psi (6 kg/cm2). Clean the filter whenever the pressure sensor triggers or coolant is replaced, and verify there are no leaks after service.；检查高压泵、过滤器和压力传感器；压力传感器触发后清洁过滤器；更换传感器后设为 85 psi (6 kg/cm²)；更换泵后确认旋向正确并至少无故障运行 10 秒；高压冷却；220 psi；压力传感器；85 psi；过滤器；冷却泵
原文摘录	9.3.1 High Pressure Coolant The TRAK high pressure coolant system, see figure 8.3.1, delivers coolant at 220 psi directly through the turret and cutting tool effectively dissipating chips and heat during machining. The increased pressure will allow for heavier cuts, higher feed rates, deep-hole drilling, extended tool life and better surface finishes. This produces reduced cycle times, increase throughput, and higher-quality parts. The high-pressure coolant pump is fitted with a filter and pressure sensor, this helps keep the filter clean and at enough pressure to ensure optimal cutting conditions. Figure 8.3.1 High Pressure Coolant 9.3.2 High Pressure Coolant Pump The high-pressure coolant system uses a larger pump delivering coolant at 220 psi. If the pump fails it will need to be replaced. Warning!!! Heavy object to be installed, proper safety precautions are needed. To replace a faulty pump: 1. Turn off fluid flow using button on the MCP and turn off power to the TCsi machine. 2. Drain enough coolant to clear the filter canister. 3. Disconnect the cable going from the electric cabinet to coolant pump and the wires at the pressure sensor, see figure 8.3.2. Note orientation of the sensor wires.

#13 lty-source-tcsi-barfeed-specification			保养维护
部件	barfeeder	章节	LNS Quick Load Servo 80 S2 Barfeed / Specifications
起始页	139	安全等级	normal
知识类型	option_specification	适用机型	TC820LTYsi

摘要 | LNS Quick Load Servo 80 S2 支持直径 6–80 mm、长度 300–1600 mm 的棒料；自动部分换料不超过 15 秒，总换料不超过 1 分钟。

A • 标识与来源

记录ID	lty-source-tcsi-barfeed-specification
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual

源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	· TC820si · TC820LTYsi
型号范围	TCsi-common
型号专属	否
章节编号	9.4.4
章节	LNS Quick Load Servo 80 S2 Barfeed / Specifications
起始页	139
结束页	139

C • 分类与治理

记录类别	maintenance
知识类型	option_specification
部件	barfeeder
安全等级	normal
可入库	是
需人工复核	否

D • 业务明细

适用机型	TC820LTYsi
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E • 检索与原文

中文摘要	LNS Quick Load Servo 80 S2 支持直径 6–80 mm、长度 300–1600 mm 的棒料；自动部分换料不超过 15 秒，总换料不超过 1 分钟。
关键词	· LNS · 棒料送料 · Barfeed · 6-80 mm · 300-1600 mm · 换料
检索文本	LNS Quick Load Servo 80 S2 支持直径 6–80 mm、长度 300–1600 mm 的棒料；自动部分换料不超过 15 秒，总换料不超过 1 分钟。；；； LNS Quick Load Servo 80 S2 specifications: diameter 1/4 to 3-1/8 in / 6 to 80 mm; bar length 14 to 63 in / 300 to 1600 mm, not exceeding the overall spindle length; loading capacity 25.5 in / 650 mm; headstock type fixed/traveling; maximum remnant length n/a; partial automatic changeover 15 seconds or less; total changeover 1 minute or less.; LNS; 棒料送料; Barfeed; 6-80 mm; 300-1600 mm; 换料
原文摘录	6. Slope setting levers Figure 8.4.3.2b Rear View 9.4.4 Specifications LNS Quick Load Servo Specifications Diameter ø 1/4" - 3-1/8" / ø 6 - 80 mm Bar Length 14" -63" / 300 mm – 1600 mm (not to exceed overall length of spindle) Loading Capacity 25.5" / 650 mm Headstock Type Fixed / traveling Max. Remnant Length n/a Partial Changeover 15 seconds or less (automatic) Total Changeover 1 minute or less